SPECIALIZATION OF THE CYCLE CLASS OF A FLAT COHERENT SHEAF OVER A VALUATION RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Relative dimension of supports of modules flat over the base (More on Morphisms, Lemma more-morphisms-lemma-relative-dimension-support-flat, canonical tag 0H3X). cited at proof lines 428-429. Dimension of the closed-fibre support is at most r given the generic-fibre bound.
- Additivity of divisors restricted to a principal vanishing order (Chow Homology, Lemma chow-lemma-additivity-divisors-restricted, canonical tag 02QG). cited at proof lines 440-446. Factors the length of the reduction modulo the uniformizer through the lengths at the minimal primes weighted by the local lengths, matching the two coefficients.

Source reference. Original lemma 0H3X: sources/stacks-chow/tags.txt, source lines 21332–21332. Cited at proof lines 428-429: the dimension bound for the closed-fibre support.

Source reference. Original lemma 02QG: sources/stacks-chow/tags.txt, source lines 3587–3587. Cited at proof lines 440-446: factors the reduction length through the minimal primes.

Lemma 0.2. *Let R be a discrete valuation ring with fraction field K and residue field κ. Let X be a scheme locally of finite type over R. Let $r \geq 0$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module flat over R. Assume $\dim(Supp(\mathcal{F}_K)) \leq r$. Then $\dim(Supp(\mathcal{F}_\kappa)) \leq r$ and*

$$sp_{X/R}([\mathcal{F}_K]_r) = [\mathcal{F}_\kappa]_r$$

Proof. The statement on dimension follows from More on Morphisms, Lemma 0H3X. Let x be a generic point of an integral closed subscheme $Z \subset X_\kappa$ of dimension r. To finish the proof we will show that the coefficient of $[Z]$ in the left (L) and right hand side (R) of equality are the same.

Let $A = \mathcal{O}_{X,x}$ and $M = \mathcal{F}_x$. Observe that M is a finite A-module flat over R. Let $\pi \in R$ be a uniformizer so that $A/\pi A = \mathcal{O}_{X_\kappa,x}$. By Chow Homology, Lemma 02QG

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we have

$$\sum\nolimits_i \text{length}_A(A/(\pi, \mathfrak{q}_i))\text{length}_{A_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) = \text{length}_A(M/\pi M)$$

where the sum is over the minimal primes $\mathfrak{q}_i$ in the support of M. Since π is a nonzerodivisor on M we see that $\pi \notin \mathfrak{q}_i$ and hence these primes correspond to those generic points $y_i \in X_K$ of the support of $\mathcal{F}_K$ which specialize to our chosen $x \in X_\kappa$. Thus the left hand side is the coefficient of $[Z]$ in (L). Of course $\text{length}_A(M/\pi M)$ is the coefficient of $[Z]$ in (R). This finishes the proof. $\square$